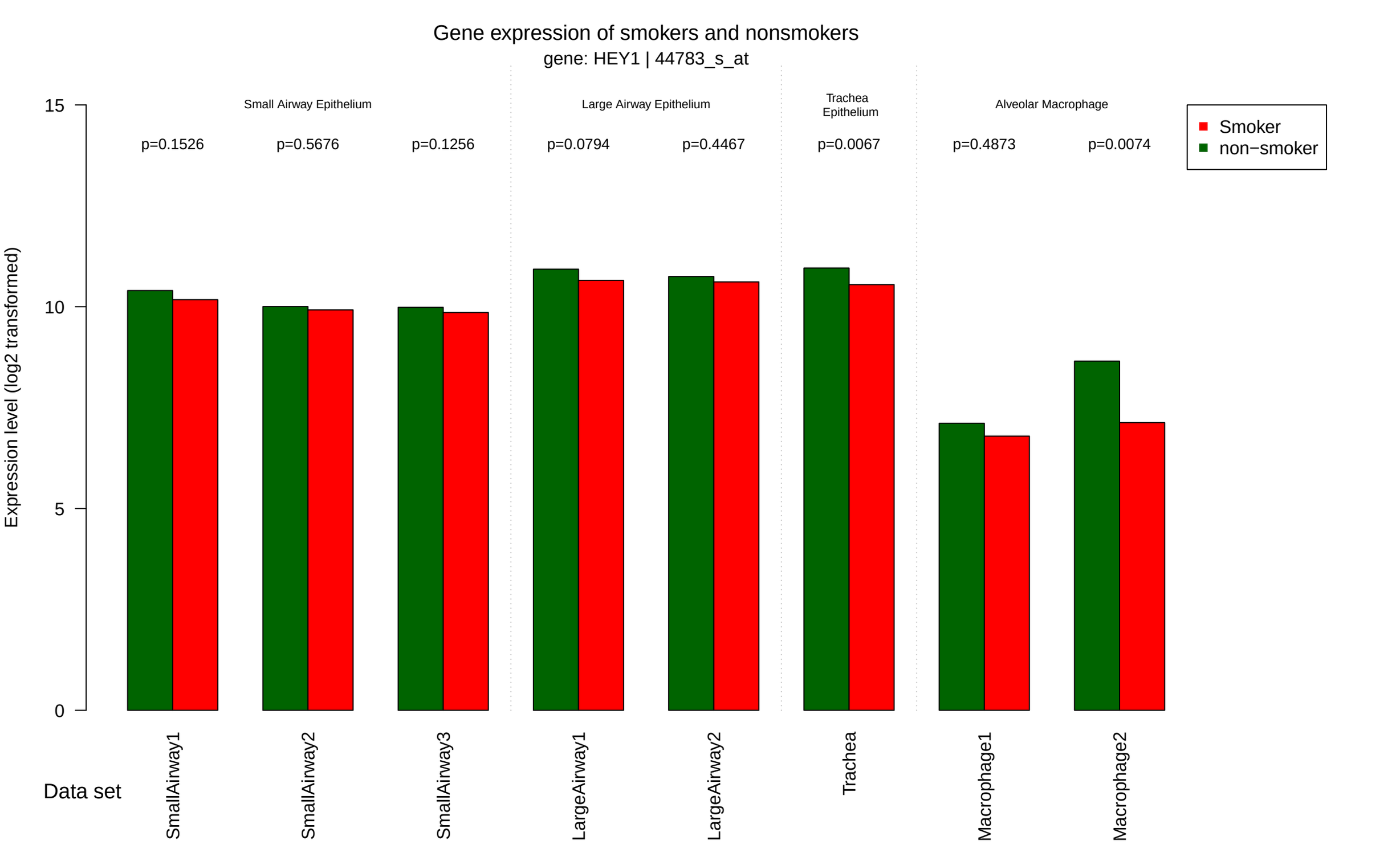
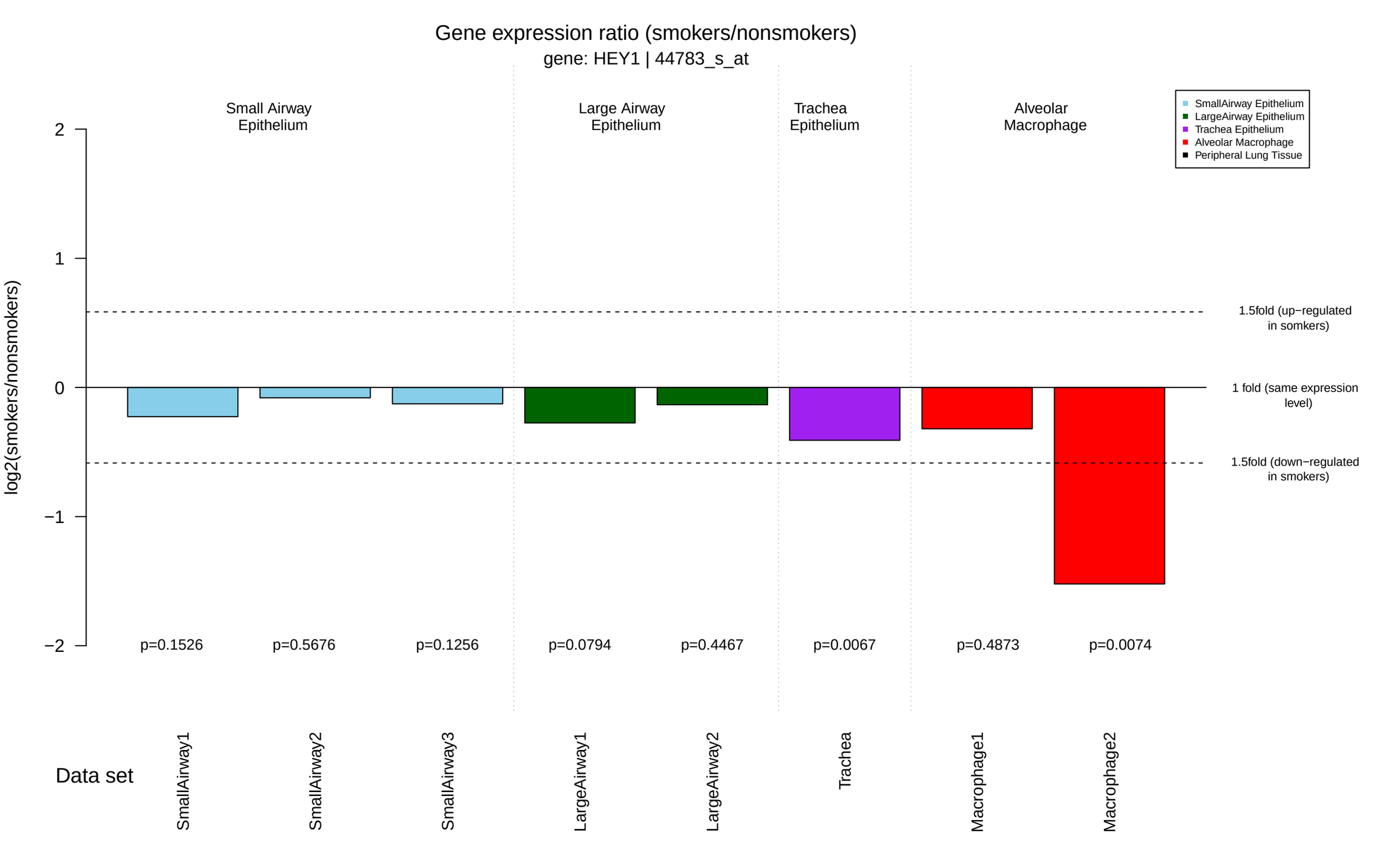
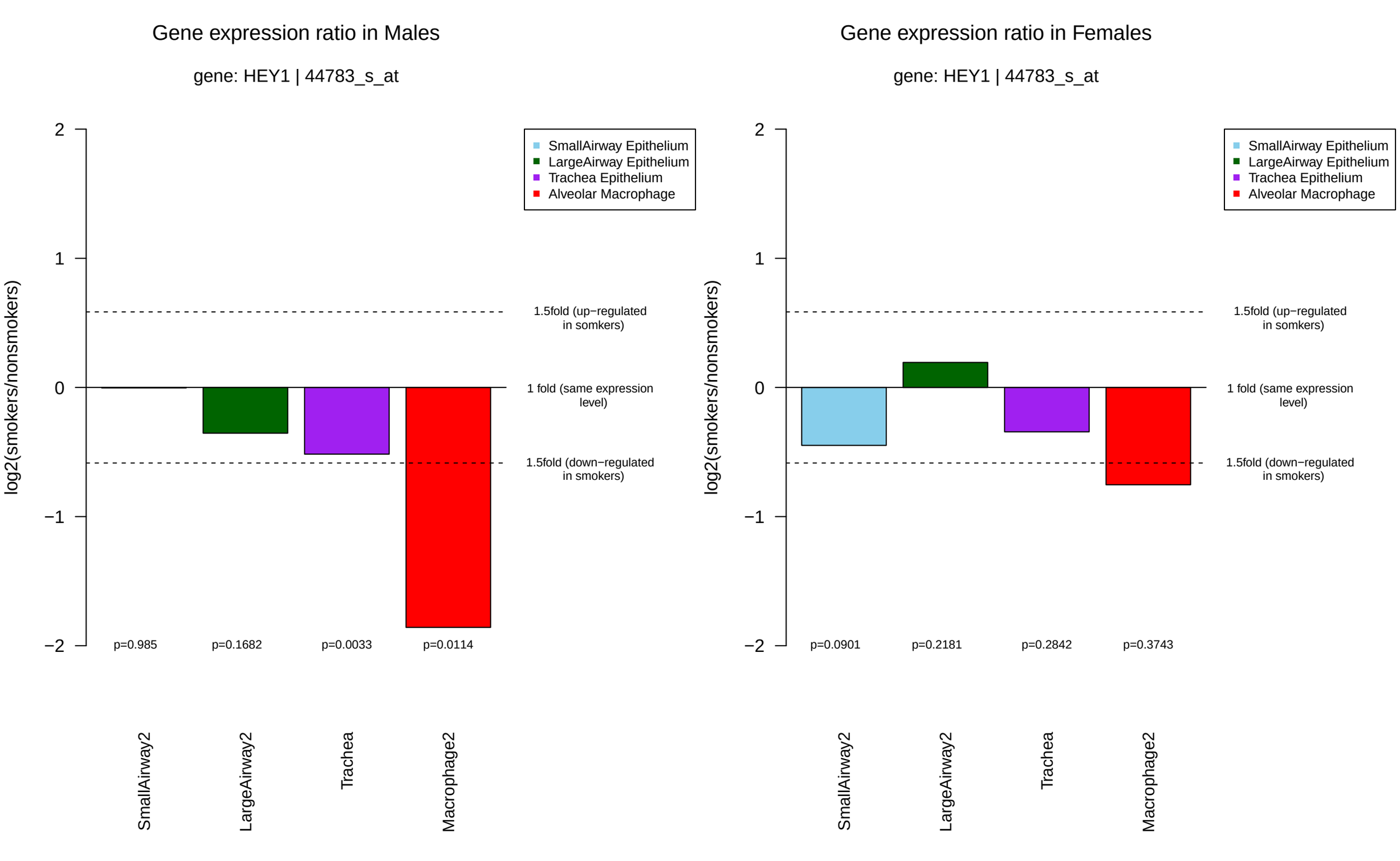
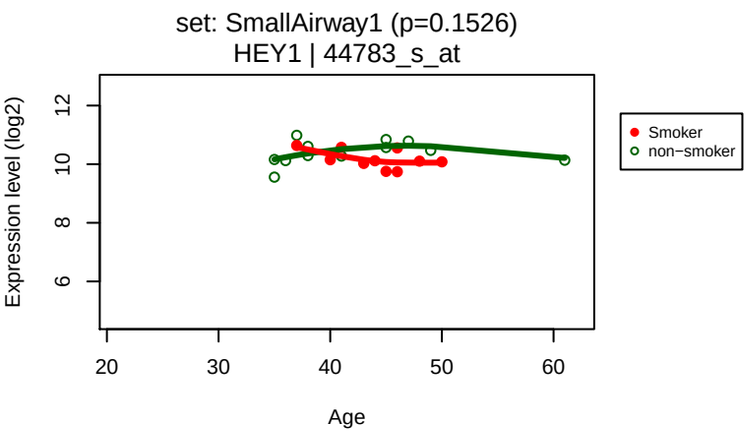
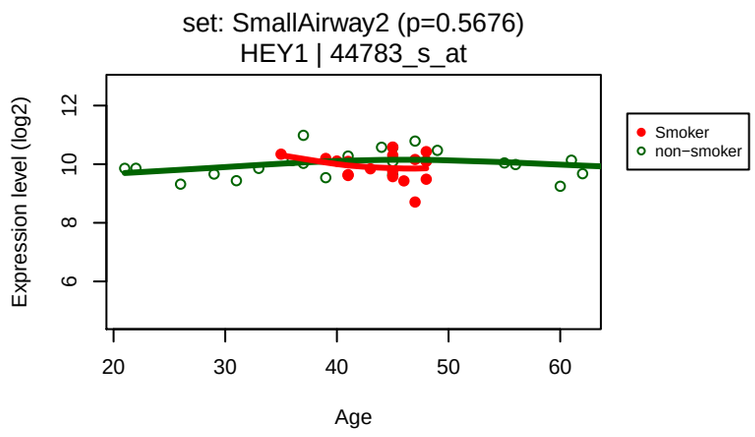
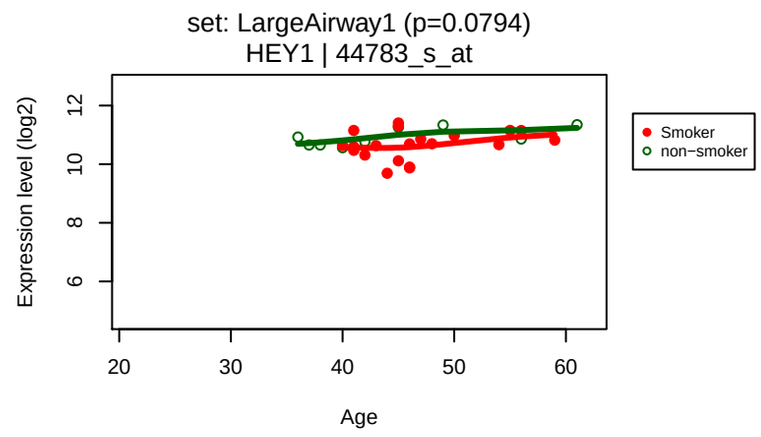

Gene expression of smokers and nonsmokers
gene: HEY1 | 44783_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1526
p=0.5676
p=0.1256
p=0.0794
p=0.4467
p=0.0067
p=0.4873
p=0.0074
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

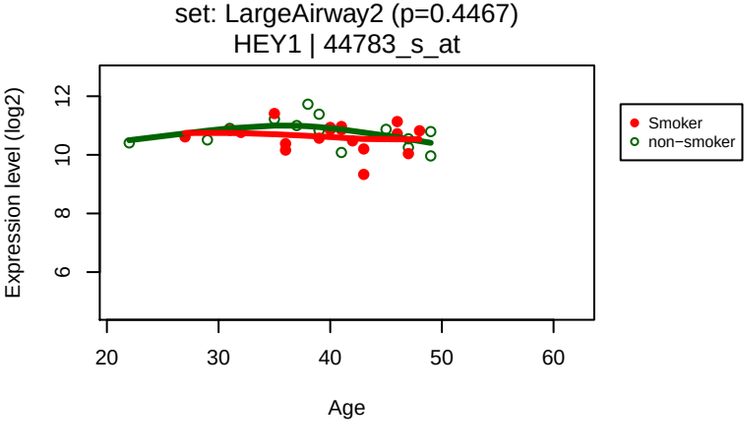

Gene expression ratio (smokers/nonsmokers)
gene: HEY1 | 44783_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1526
p=0.5676
p=0.1256
p=0.0794
p=0.4467
p=0.0067
p=0.4873
p=0.0074
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

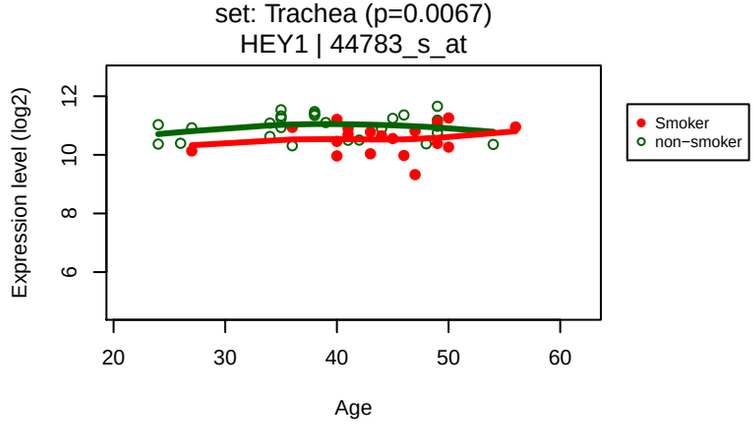

Gene expression ratio in Males
gene: HEY1 | 44783_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.985
p=0.1682
p=0.0033
p=0.0114
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HEY1 | 44783_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0901
p=0.2181
p=0.2842
p=0.3743
SmallAirway2
LargeAirway2
Trachea
Macrophage2

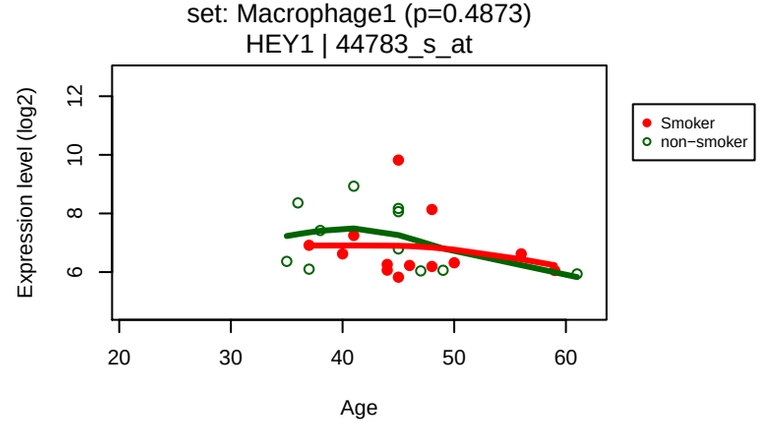

set: SmallAirway1 (p=0.1526)
HEY1 | 44783_s_at
Expression level (log2)
Age
Smoker
non−smoker

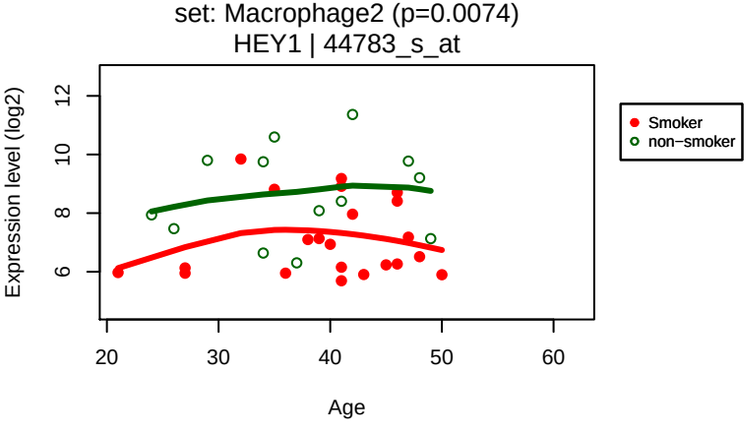

set: SmallAirway2 (p=0.5676)
HEY1 | 44783_s_at
Expression level (log2)
Age
Smoker
non−smoker

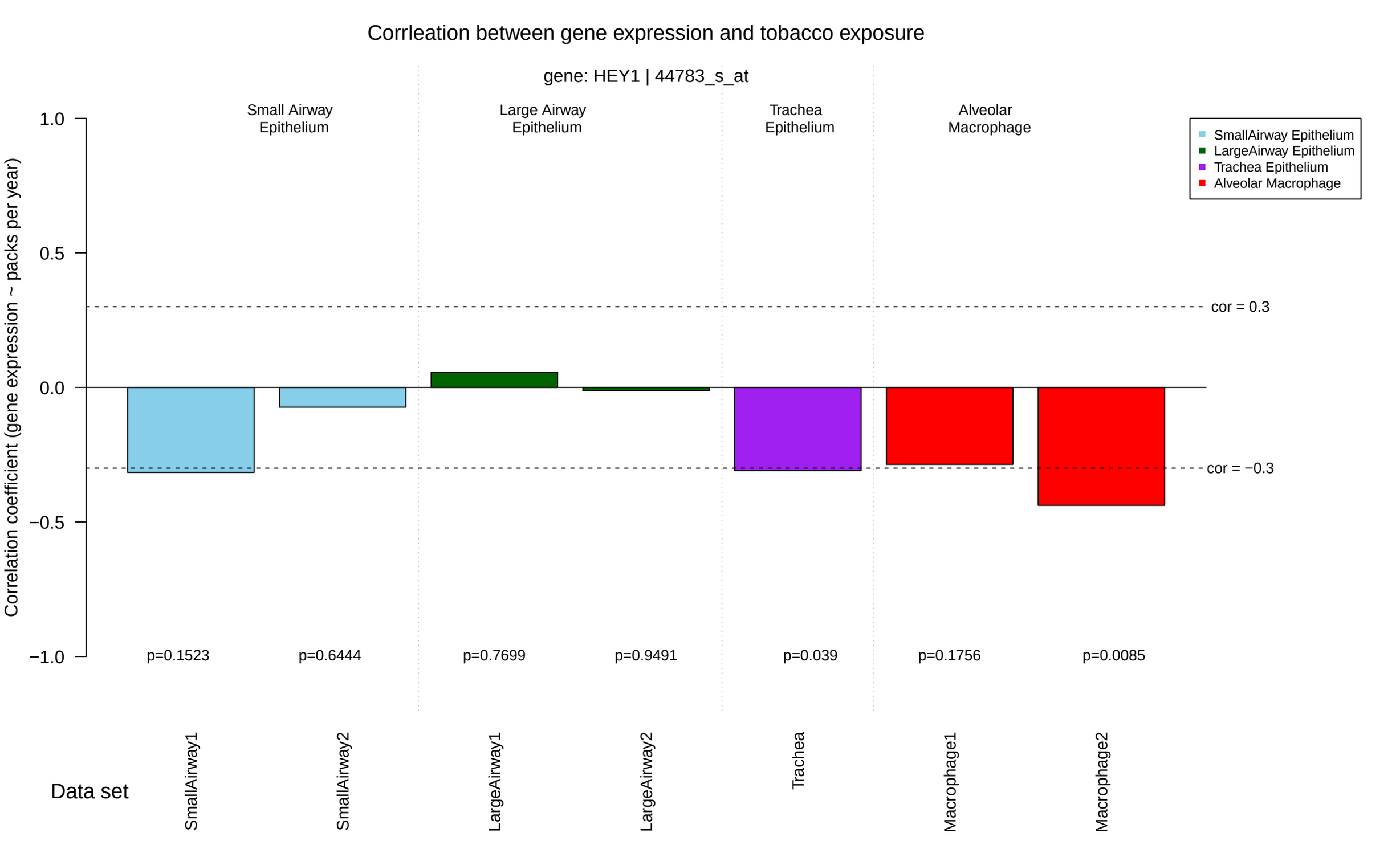

set: LargeAirway1 (p=0.0794)
HEY1 | 44783_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4467)
HEY1 | 44783_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0067)
HEY1 | 44783_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4873)
HEY1 | 44783_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0074)
HEY1 | 44783_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HEY1 | 44783_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1523
p=0.6444
p=0.7699
p=0.9491
p=0.039
p=0.1756
p=0.0085
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2